# A unique-minimizer nonconvergence example for undamped adaptive Bregman proximal gradient

**Abstract**

We construct a two-dimensional Legendre kernel and a globally smooth convex objective with a unique minimizer for which the literal undamped ambient B-adaPG controller generates a nonconvergent orbit. The kernel is finite and continuous on a closed rectangle, has positive definite Hessian in its interior, and satisfies the full boundary Bregman-zone assumptions. The objective is $x^2 + y$. An explicit infinite construction of its conjugate forces infinitely many exact excursions to first coordinate $1/2$, each followed by an exact return to zero. Every stepsize is selected by the original controller from a specified positive initial pair. All forward probes, joins, and tail dependencies are verified. The example concerns compact-interior relative smoothness; boundary-uniform relative smoothness fails. It is compatible with a unique optimal cluster point and makes no claim about a damped rule or an uninspected journal formulation. No novelty claim is made.

**Keywords:** adaptive Bregman proximal gradient; unique minimizer; nonconvergent orbit; Legendre kernel; boundary relative smoothness; convex counterexample

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## 1 The precise statement and controller

We use the undamped B-adaPG rule in equations (2.3) and (2.6) of the pinned arXiv version 2 of Ou, Latafat, and Themelis [1]. The composite term is $g = 0$. Uppercase $X = (x, y)$ denotes a primal point; the letter $g_j$ below will denote a positive incoming stepsize, not the composite term. For a differentiable convex function $v$, write

$$D_v(U, V) = v(U) - v(V) - \langle \nabla v(V), U - V \rangle, \qquad \Delta_v(U, V) = D_v(U, V) + D_v(V, U).$$

At a boundary first argument, the same definition is used whenever $v$ is finite there and the second argument is interior.

Let $p_k = \nabla h(X_k)$. The first update uses $\gamma_1$, and every update is

$$X_{k+1} = \nabla h^*(p_k - \gamma_{k+1} \nabla f(X_k)). \tag{1}$$

For $k \geq 1$, given distinct consecutive points, the literal controller is

$$\begin{aligned}
\rho_k &= \gamma_k/\gamma_{k-1}, \quad r_{k+1} = \sqrt{1+\rho_k}, \quad \delta_{k+1} = 2r_{k+1}, \\
H_k &= \nabla h - \gamma_k \nabla f, \\
\ell_k &= \frac{\Delta_f(X_k, X_{k-1})}{\Delta_h(X_k, X_{k-1})}, \quad \alpha_k = \frac{D_h(X_k, X_{k-1})}{D_h(X_{k-1}, X_k)}, \\
\Lambda_k &= \frac{2D_{h^*}\big(p_k + \delta_{k+1}[H_k(X_k) - H_k(X_{k-1})], p_k\big)}{\delta_{k+1}^2 \Delta_h(X_k, X_{k-1})}, \\
\rho_{k+1} &= \min\left\{ r_{k+1}, \frac{\alpha_k}{(1+\alpha_k)2r_{k+1}[\Lambda_k - (1-\gamma_k\ell_k)]_+} \right\}, \\
\gamma_{k+1} &= \rho_{k+1}\gamma_k.
\end{aligned} \tag{2}$$

Positive division by zero means $+\infty$. Repeated primal iterates would be stopped before evaluating zero denominators. We prove that this never occurs.

**Theorem 1.** *There exist $e > 0$, a proper closed convex 1-coercive Legendre function $h$, and $X_0 \in C$, where*

$$K = \operatorname{dom} h = [-1, 1] \times [0, e], \qquad C = \operatorname{int} K,$$

*with the following properties. The function $h$ is finite and continuous on $K$, is $C^2$ on $C$, and has positive definite Hessian there. For every $Z \in K$ and every sequence $C \ni Y_i \to Z$,*

$$D_h(Z, Y_i) \longrightarrow 0.$$

*For each $Z \in K \setminus C$, every sublevel of $D_h(Z, \cdot)$ in $C$ is bounded. Thus the full source Assumption 2.4 holds.*

*The function $f(x, y) = x^2 + y$, with $g = 0$, satisfies the source compact-interior local relative smoothness assumption and has the unique minimizer $(0, 0)$ on $K$. Nevertheless, the initialized rule (1)–(2), with*

$$\phi = \frac{1+\sqrt{5}}{2}, \qquad \gamma_0 = \phi^{-2}, \qquad \gamma_1 = \phi^{-1},$$

*has subsequences converging to $(0, 0)$ and $(1/2, 0)$. All its points and evaluations are well-defined, all stepsizes are finite and positive, and it never stops. In particular the orbit does not converge, while*

$$\liminf_k f(X_k) = 0, \qquad \limsup_k f(X_k) \geq \frac{1}{4}.$$

The proof below constructs a single fixed ambient $h^*$, not a restricted conjugate, a projection, or assigned local Hessians. Neither later stepsizes nor the spikes are externally prescribed to the algorithm. The recursive parameters define the kernel; exact controller identities then force the claimed orbit.

## 2 Dual coordinates and the local transition

Set

$$c = \frac{1}{2}, \qquad v = \operatorname{atanh}(c) = \frac{1}{2}\log 3, \qquad B(z) = \log\cosh z, \qquad b = \frac{B(v)}{v}.$$

The cutoff used for every join is

$$S(u) = \begin{cases} 0, & u \le 0, \\ 10u^3 - 15u^4 + 6u^5, & 0 \le u \le 1, \\ 1, & u \ge 1. \end{cases} \tag{3}$$

It is $C^2$, lies in $[0,1]$, is nondecreasing, and its first two derivatives vanish at $0, 1$. Moreover $\max|S''| = 10\sqrt{3}/3 < 6$. The dual potential will have the form

$$\psi(s,t) = Q(t) + \varepsilon(t) B\left(\frac{s - a(t)}{\varepsilon(t)}\right), \qquad h^*(p_x, p_y) = \psi(p_x, -p_y). \tag{4}$$

Thus $p = (s, -t)$ and $X = (\psi_s, -\psi_t)$. The objective gradient is $(2x, 1)$, so (1) becomes exactly

$$(s,t) \longmapsto (s - 2\gamma x, t + \gamma). \tag{5}$$

The previous update also gives the useful identity

$$H_k(X_k) - H_k(X_{k-1}) = -\gamma_k \nabla f(X_k). \tag{6}$$

The ambient probe therefore advances by $(-4r\gamma_k x, 2r\gamma_k)$ in $(s,t)$ coordinates. Conjugate Bregman distances reverse orientation: $D_h(U,V) = D_{h^*}(\nabla h(V), \nabla h(U))$.

Consider a transition beginning at $t = T$, with incoming centered step $g > 0$ and incoming ratio $\rho \in (1, \phi]$. Define

$$r = \sqrt{1+\rho}, \qquad w = rg, \qquad R = \sqrt{1+r}, \qquad 0 < \theta = \frac{A}{w} \le 10^{-6}.$$

In particular $1 \le r \le \phi$ and $\sqrt{2} \le R \le \phi$. Set

$$\begin{aligned} q &= \frac{4Rwc}{A}, \\ D_{\tan} &= \frac{A}{v}\{B(v(1-q)) - B(v)\} + 4Rwc^2, \\ C_0 &= 2b + \frac{D_{\tan}}{Rcw} + 4Rc, \\ \lambda &= \frac{C_0 A}{w^2}, \qquad \varepsilon = \frac{A}{v}. \end{aligned} \tag{7}$$

Here $C_0$ is a scalar coefficient, unrelated to the primal open set $C$. On the transition and its needed probes use

$$a(t) = a_{\text{old}} - AS((t-T)/w), \qquad \varepsilon(t) = A/v, \qquad Q''(t) = \lambda. \tag{8}$$

A centered starting point $(a_{\text{old}}, T)$ has $x = 0$. A step $w$ reaches $(a_{\text{old}}, T + w)$, where $x = \tanh v = c$ and $a' = 0$. Its $y$ coordinate is $-Q'$.

### 2.1 Uniform exact coefficient bounds

For $q > 1$, put $\eta = \log(1 + \exp(-2v(q-1)))$, so $0 < \eta < \log 2$. The exact logarithm identity yields

$$\begin{aligned} D_{\tan} &= 4Rwc(1+c) - Ad, \\ d &= 1 + b + (\log 2 - \eta)/v, \\ C_0 &= 2b + 4(1 + c + Rc) - \frac{\theta d}{Rc}. \end{aligned} \tag{9}$$

For clarity, rational loose bounds suffice throughout: $1/4 < b < 27/100$, $1/2 < v$, $\log 2 < 7/10$, and therefore $0 < d < 267/100$. These logarithmic inequalities follow, for example, by comparing powers of 2 and 3, using $b = 2\log 2/\log 3 - 1$. Also $7/5 < \sqrt{2}$ and $\phi < 13/8$. Hence the limiting expression in (9) lies strictly between $93/10$ and $979/100$. Since $Rc \geq 1/\sqrt{2} > 7/10$, the subtracted term is less than $4 \cdot 10^{-6}$. Consequently

$$9 < C_0 < 10. \tag{10}$$

In particular $C_0 > 2\phi^2 b$ and $C_0 > 2\phi^4 b$: the latter follows already from $2(13/8)^4(27/100) < 4 < 9$. No numerical trajectory or fitted root is involved.

## 3 Four exact controller verifications

### 3.1 Departure to the spike

Assume the previous centered edge has length $g = w/r$ and lies in the constant $\lambda, a, \varepsilon$ plateau. Then

$$\ell = 0, \qquad \alpha = 1, \qquad \Delta_h = \lambda g^2.$$

By (6), the probe from $(a_{\text{old}}, T)$ advances normally by $2w$. The decrease of $a$ has finished at its endpoint, whence

$$D_\psi(\text{probe}, \text{current}) = 2\lambda w^2 + Ab, \qquad \Lambda = 1 + \frac{b}{2C_0}.$$

For $\alpha = 1$, the growth candidate $r$ is selected precisely when $\Lambda - 1 \leq 1/(4r^2)$. This follows strictly from $C_0 > 2r^2 b$. Thus the actual next stepsize is $w$ and its endpoint has $x = c$.

### 3.2 The spike forces an exact return

For the departure edge, direct evaluation with reversed dual orientation gives

$$\begin{aligned} d_h &:= D_h(X_{\text{spike}}, X_{\text{center}}) = A(C_0/2 - b), \\ \bar{d}_h &:= D_h(X_{\text{center}}, X_{\text{spike}}) = A(C_0/2 + b), \\ \Delta_h &= C_0 A, \qquad \Delta_f = 2c^2. \end{aligned} \tag{11}$$

The new growth candidate is $R$. Its ambient probe is

$$(a_{\text{old}} - 4Rwc,\ T + w + 2Rw).$$

At its endpoint the baseline is $a_{\text{new}} = a_{\text{old}} - A$. The tangential Bregman term is exactly $D_{\tan}$ of (7), and

$$D_\psi(\text{probe}, \text{spike}) = 2R^2 C_0 A + D_{\tan}.$$

This tangential term is nonnegative by convexity of $s \mapsto \varepsilon B((s - a_{\text{new}})/\varepsilon)$. The normal quadratic contribution cancels in the controller bracket:

$$[\Lambda - 1 + w\ell]\Delta_h = \frac{D_{\tan}}{2R^2} + 2wc^2 > 0. \tag{12}$$

Since $\alpha/(1+\alpha) = d_h/\Delta_h$, the second candidate for the next stepsize is

$$\frac{wA(C_0/2 - b)}{D_{\tan}/R + 4Rwc^2} = \frac{A}{2c} = A. \tag{13}$$

The equality is exactly the defining equation for $C_0$. As $A < Rw$, this candidate is selected. Applying (5) returns exactly to

$$(a_{\text{old}} - A,\ T + w + A) = (a_{\text{new}},\ T + w + A),$$

with $x = 0$.

### 3.3 Growth after the return and along centered plateaus

Suppose $a = a_{\text{new}}$ after the return and $Q''$ is nonincreasing in the subsequent interval under consideration. At the return edge,

$$\Delta_h = Ac + A\int_{T+w}^{T+w+A} Q''(u)\,du, \qquad A\Delta_f = 2Ac^2 = Ac. \tag{14}$$

At its centered endpoint the probe is purely normal. If its growth parameter is $r'$, then the probe length is $2r'A$. On this centered ray $\psi(a_{\text{new}}, t) = Q(t)$, independently of later changes of $\varepsilon$. The integral formula for $D_Q$ and monotonicity give

$$\Lambda\Delta_h \le A^2 Q''(T + w + A) \le A\int_{T+w}^{T+w+A} Q''(u)\,du.$$

Together with (14), this says $\Lambda - (1 - A\ell) \le 0$. The second branch is infinite and growth is selected.

More generally, on a wholly centered edge of length $\gamma$ ending at $t$, whose probe stays on the same baseline,

$$\Delta_h = \gamma\int_{t-\gamma}^{t} Q''(u)\,du, \qquad \ell = 0,$$

while $2D_Q(t + 2r\gamma, t)/(2r)^2 \le \gamma^2 Q''(t) \le \Delta_h$. Hence $\Lambda \le 1$, so growth is selected. This argument permits changes in $\varepsilon$ and decreases of $Q''$; it does not assume $\alpha = 1$ during such changes.

### 3.4 The preceding probe that sees a future transition

One earlier controller may look into the next transition. At $T-g$, let the current incoming step be $g/u$, and let its growth candidate be $u \le \phi$. Assume its old edge and probe have constant $Q'' = \lambda$ and $\varepsilon = A/v$. Its probe ends at $T+g$, whose cutoff argument is $1/r$. Therefore

$$\Lambda = 1 + \frac{r^2 b_S}{2C_0}, \qquad b_S = \frac{B(vS(1/r))}{v} \le b.$$

The old edge is centered with constant curvature, so $\alpha = 1$ and $\ell = 0$. Growth is selected if $C_0 \ge 2u^2 r^2 b_S$, which follows strictly from $C_0 > 2\phi^4 b$. All still earlier centered probes end before $T$. Indeed, for increasing centered steps $\Gamma_n$, the probe from $t_n$ ends at $t_n + 2\Gamma_{n+1}$, and

$$t_n + 2\Gamma_{n+1} < t_n + \Gamma_{n+1} + \Gamma_{n+2} = t_{n+2}. \tag{15}$$

If $t_m = T$, this excludes every probe with $n \le m-2$. Thus the crossing probe just checked is the only additional one.

## 4 A fully specified infinite construction

### 4.1 Recursive parameters and placement

Set $T_1 = 0$, $w_1 = 1$, and $g_1 = \phi^{-1}$. The first incoming ratio is $\phi$ and its departure growth candidate is $r_1 = \phi$, so $w_1 = r_1 g_1$. At stage $j$, the numbers $T_j, w_j$ and the incoming ratio have already been determined. Define

$$A_j = \min\left\{10^{-7} w_j,\ \frac{2^{-j} w_j^2}{20(T_j + 60w_j + 1)^2},\ \frac{A_{j-1}}{4},\ \frac{\lambda_{j-1} w_j^2}{40}\right\}, \tag{16}$$

where the last two entries are omitted for $j = 1$. With $A = A_j$, $w = w_j$, and its known growth candidate $r = r_j$, define $C_j$ by (7) and set

$$\lambda_j = C_j A_j / w_j^2, \qquad \varepsilon_j = A_j / v.$$

These are exact positive real numbers. The bounds just proved imply

$$\begin{gathered} A_j/w_j \le 10^{-7}, \qquad A_j \le A_{j-1}/4, \qquad \lambda_j \le \lambda_{j-1}/4 \quad (j > 1), \\ \lambda_j (T_j + 60w_j + 1)^2 \le 2^{-j}/2. \end{gathered} \tag{17}$$

After the return from transition $j$, form the prospective centered growth sequence

$$\begin{aligned} \Gamma_{-1} &= w_j, \qquad \Gamma_0 = A_j, \\ \Gamma_{n+1} &= \Gamma_n \sqrt{1 + \Gamma_n/\Gamma_{n-1}} \quad (n \ge 0), \\ t_n &= T_j + w_j + A_j + \sum_{i=1}^{n} \Gamma_i \quad (n \ge 0). \end{aligned} \tag{18}$$

Choose the least integer $m \geq 2$ satisfying all three conditions

$$\begin{aligned} \Gamma_m &\geq 1000 w_j, \\ r_{\text{next}} := \sqrt{1 + \Gamma_m/\Gamma_{m-1}} &\geq 8/5, \\ t_m - \Gamma_m - \Gamma_{m-1} &\geq T_j + 100 w_j. \end{aligned} \tag{19}$$

Then set

$$T_{j+1} = t_m, \quad g_{j+1} = \Gamma_m, \quad r_{j+1} = r_{\text{next}}, \quad w_{j+1} = r_{\text{next}} \Gamma_m. \tag{20}$$

The incoming ratio at the new stage is $\Gamma_m/\Gamma_{m-1}$.

This recursion terminates at every stage. Indeed the step ratios obey $q_{n+1} = \sqrt{1+q_n}$, starting from $q_0 = A_j/w_j < 1$. They increase strictly to $\phi$, with $q_1 > 1$ and $q_n > \sqrt{2}$ for $n \geq 2$. Thus $\Gamma_n \to \infty$ and $t_{n-2} \to \infty$, so all conditions in (19) eventually hold. All ratios used in the transition checks are in $(1, \phi]$. Also

$$w_{j+1} \geq 1600 w_j, \qquad T_{j+1} > T_j + 100 w_j. \tag{21}$$

There is no finite accumulation of transitions.

## 4.2 Global baseline, width, and curvature joins

The geometric bound on $A_j$ makes $a_0 = \sum_{j \geq 1} A_j$ finite. Define on all of $\mathbb{R}$

$$a(t) = a_0 - \sum_{j \geq 1} A_j S((t - T_j)/w_j). \tag{22}$$

The transition supports are disjoint and locally finite. Thus $a$ is nonnegative and $C^2$, is constant $a_0$ before $T_1$, and tends to zero as $t \to +\infty$.

Set $\varepsilon = \varepsilon_1$ before the first width join. For each $j$, leave $\varepsilon = \varepsilon_j$ through $T_j + 20w_j$, and on $[T_j + 20w_j, T_j + 30w_j]$ set

$$\varepsilon(t) = \varepsilon_j + (\varepsilon_{j+1} - \varepsilon_j) S\left(\frac{t - T_j - 20w_j}{10 w_j}\right). \tag{23}$$

After this join leave $\varepsilon = \varepsilon_{j+1}$ until the next width join. Separation (21) makes these instructions consistent. The result is positive and $C^2$ at every finite $t$, bounded, and tends to zero at $+\infty$. It is constant on the negative tail.

Define a positive continuous curvature function $q_0$ as follows:

$$q_0(t) = \lambda_1 e^{t+g_1} \quad (t < -g_1), \qquad q_0(t) = \lambda_1 \quad (-g_1 \leq t \leq T_1 + 40w_1).$$

For each $j$, keep the value $\lambda_j$ through $T_j + 40w_j$, set

$$q_0(t) = \lambda_j + (\lambda_{j+1} - \lambda_j) S\left(\frac{t - T_j - 40w_j}{10 w_j}\right) \tag{24}$$

on $[T_j + 40w_j, T_j + 50w_j]$, and then keep $\lambda_{j+1}$ until the next curvature join. The exponential join at $-g_1$ is continuous; differentiability of $q_0$ there is unnecessary. In particular $q_0$ is nonincreasing on $[-g_1, \infty)$. Finally define

$$Q(t) = \int_t^\infty (u - t) q_0(u)\, du, \qquad e = \int_{\mathbb{R}} q_0(u)\, du. \tag{25}$$

## 4.3 Finite first moment and differentiability

Partition $[0,\infty)$ into the first interval $[0, T_1 + 50w_1]$ and, for $j \geq 2$, the intervals

$$I_j = [T_{j-1} + 50w_{j-1},\ T_j + 50w_j].$$

On the $j$th interval $q_0 \leq \lambda_j$, including its final downward join. Since all endpoints are nonnegative,

$$\int_{I_j} (1+t)q_0(t)\,dt \leq \lambda_j (T_j + 50w_j + 1)^2 \leq 2^{-j}/2,$$

with the same estimate for the first interval. The negative exponential tail has finite mass and finite first moment as well. Therefore

$$0 < e < \infty, \qquad \int_{\mathbb{R}} (1+|t|)q_0(t)\,dt < \infty. \tag{26}$$

It follows by differentiation of (25) that $Q$ is finite and $C^2$, $Q'' = q_0 > 0$, $Q'(t) = -\int_t^\infty q_0$, and

$$Q(t), Q'(t) \longrightarrow 0 \quad (t \to +\infty).$$

The derivative amplitudes of the $a$ and $\varepsilon$ joins are bounded by constant multiples of $A_j/w_j$. This tends to zero, since $A_j$ decreases geometrically and $w_j$ increases. Thus $a', \varepsilon' \to 0$ at $+\infty$, and both derivatives are identically zero sufficiently far on the negative tail. The functions $a, \varepsilon$ themselves tend to zero only at $+\infty$, not at both ends.

## 4.4 All buffers and dependence on the future tail

A transition's spike probe ends before $T_j + 5w_j$, because $1 + 2R < 5$; its departure and post-return probes are shorter. Thus they lie before the width join at $+20w_j$ and curvature join at $+40w_j$. The constant-profile assumptions in the three local checks hold. The last placement condition says $t_{m-2} \geq T_j + 100w_j$. Hence both incoming centered edges for the next transition begin after the preceding width and curvature joins, which end by $T_j + 50w_j$. They see exactly $\varepsilon_{j+1}$ and $\lambda_{j+1}$, as required for the departure check and the single preceding crossing probe. All other centered probes stay before the next baseline change by (15). They use the monotone-curvature argument. This proves that every prospective step in (18) is the step actually selected by the controller.

There is no circular choice of parameters: $A_j, C_j, \lambda_j$ are chosen from known stage-$j$ data; the scalar recurrence then chooses $T_{j+1}, w_{j+1}$; only afterward is $A_{j+1}$ chosen. That later choice fixes the endpoint of an already scheduled width or curvature join. Its value cannot spoil any earlier centered-growth inequality. More generally, altering $q_0$ strictly to the right of an old finite interval changes $Q$ on that interval only by an affine function. Dual Bregman divergences are invariant under that change. The induced common translation of primal $y$ values changes neither $\nabla f = (2x, 1)$ nor $\Delta_f = 2(\Delta x)^2$. Likewise a future summand in $a_0$ translates all earlier $s$ and $a$ by the same amount, leaving $s - a$ and every

local calculation unchanged. The infinite series and integrals first define one fixed kernel and one fixed initial point; the preceding verifications are valid for its finite prefixes. They do not modify an instance while it is running.

# 5 Global convexity and the ambient Legendre kernel

## 5.1 Positive definite dual Hessian

Use the globally defined functions in (4). Put $z = (s - a(t))/\varepsilon(t)$ and $E(z) = B(z) - zB'(z)$. Then

$$B''(z) > 0, \qquad |B'(z)| < 1, \qquad -\log 2 \le E(z) \le 0.$$

Direct differentiation gives $\psi_{ss} = B''(z)/\varepsilon > 0$ and the exact Hessian Schur complement

$$\psi_{tt} - \frac{\psi_{st}^2}{\psi_{ss}} = Q''(t) + \varepsilon''(t)E(z) - a''(t)B'(z). \tag{27}$$

Indeed the terms quadratic in $a' + z\varepsilon'$ cancel in this expression. On an $a$-transition, $|a''| < 6A_j/w_j^2$ while $Q'' = C_jA_j/w_j^2 > 9A_j/w_j^2$ and $\varepsilon'' = 0$. On its disjoint width join,

$$(\log 2)|\varepsilon''| \le \frac{6(\log 2)A_j}{100vw_j^2} < \frac{A_j}{w_j^2} < Q'',$$

and $a'' = 0$. Elsewhere both second derivatives vanish. Thus globally

$$Q'' > |a''| + (\log 2)|\varepsilon''|. \tag{28}$$

The joins are $C^2$ and locally finite, so $\psi$ and $h^*$ are finite $C^2$ functions on all of $\mathbb{R}^2$ with continuous positive definite Hessian. In particular they are strictly convex.

## 5.2 Strict interior image and exact conjugate domain

Integrate (28) toward $+\infty$ and $-\infty$, using the derivative limits already established. This gives, for every finite $t$,

$$\begin{aligned} -Q'(t) &> |a'(t)| + (\log 2)|\varepsilon'(t)|, \\ e + Q'(t) &> |a'(t)| + (\log 2)|\varepsilon'(t)|. \end{aligned} \tag{29}$$

Strictness holds because the continuous positive difference in (28) has positive integral on each nonempty tail. Since

$$\psi_s = B'(z), \qquad \psi_t = Q' + \varepsilon' E(z) - a'B'(z),$$

we obtain

$$\nabla h^*(\mathbb{R}^2) \subset (-1, 1) \times (0, e) = C. \tag{30}$$

The elementary estimate

$$|\varepsilon B((s-a)/\varepsilon) - |s|| \le |a| + \varepsilon \log 2$$

is uniform, since $a, \varepsilon$ are bounded. Also

$$Q(t) = \int_{\mathbb{R}} (u-t)_+ q_0(u)\, du, \quad |Q(t) - e(-t)_+| \le \int_{\mathbb{R}} |u| q_0(u)\, du.$$

Consequently $h^*(p)$ differs by a bounded amount from the support function $\sigma_K(p) = |p_x| + e(p_y)_+$ of $K = [-1,1] \times [0,e]$. Define $h = (h^*)^*$. Conjugating the uniform support-function bounds gives, for some finite $M$,

$$\iota_K - M \le h \le \iota_K + M. \tag{31}$$

Thus $h$ is proper closed convex, finite and bounded on every point of $K$, and infinite outside $K$. Its bounded domain implies 1-coercivity in the extended-real sense. Since $h^*$ is finite closed convex, $h^*$ is indeed the full ambient conjugate of this $h$.

For every $X \in C$, there is a positive support-function margin $\delta_X$ such that $\sigma_K(p) - \langle p, X\rangle \ge \delta_X \|p\|$. Hence $h^*(p) - \langle p, X\rangle$ is coercive. It attains a unique minimum, whose first-order equation is $\nabla h^*(p) = X$. Strict convexity makes the gradient injective, so its image is exactly $C$. The inverse function theorem and the positive definite dual Hessian now imply

$$h \in C^2(C), \qquad \nabla h = (\nabla h^*)^{-1}, \qquad \nabla^2 h(X) = \left[\nabla^2 h^*(\nabla h(X))\right]^{-1} \succ 0.$$

No boundary point $X \in K \setminus C$ has a subgradient of $h$: otherwise Fenchel reciprocity would put $X$ in the image of $\nabla h^*$, contradicting (30). If $X_i \in C$ approaches the boundary and $\nabla h(X_i)$ has a bounded subsequence, continuity of $\nabla h^*$ would likewise produce a boundary image of a finite dual point. Thus the gradient norm diverges on every such approach. This proves essential smoothness. The positive interior Hessian and $\operatorname{dom} \partial h = C$ imply essential strict convexity, so $h$ is Legendre.

## 5.3 Continuity on every face and the full boundary assumption

The bound (31) alone does not assert boundary continuity. Here it follows from the rectangular geometry. Fix $Z \in K$. Choose $\delta_Z > 0$ smaller than every positive slack of the four normalized affine inequalities defining $K$. Every feasible unit ray direction $d$ from $Z$ then satisfies $Z + \delta_Z d \in K$: a zero-slack inequality imposes an inward sign, and each positive slack permits this common distance. For $X \in K$ with $0 < \|X - Z\| < \delta_Z$, let $\tau = \|X - Z\|/\delta_Z$ and $d = (X - Z)/\|X - Z\|$. Convexity and the uniform upper bound give

$$h(X) \le (1-\tau)h(Z) + \tau h(Z + \delta_Z d) \le (1-\tau)h(Z) + \tau M.$$

This proves upper semicontinuity at $Z$; closedness supplies lower semicontinuity. Therefore $h$ is continuous on all of $K$, including its corners. For all sufficiently nearby $X \in C$, the same slack argument gives $2X - Z \in K$. Convexity yields the two-sided estimate

$$h(X) - h(2X - Z) \leq \langle \nabla h(X), Z - X \rangle \leq h(Z) - h(X).$$

Both ends tend to zero as $X \to Z$. Thus $D_h(Z, X) \to 0$ for every $Z \in K$ and every interior approach. Every reverse-divergence sublevel in $C$ is bounded because $C \subset K$ is bounded. This verifies both parts of source Assumption 2.4 on the whole domain, not just on the constructed orbit.

# 6 Initialization, actual orbit, and scope

The globally fixed initial point is

$$X_0 = \nabla h^*(a_0, g_1), \qquad \gamma_0 = g_1/\phi = \phi^{-2}, \qquad \gamma_1 = g_1 = \phi^{-1}. \tag{32}$$

In transformed coordinates this is $(s, t) = (a_0, -g_1)$ and is centered. The first primal update uses $\gamma_1$ and reaches $(a_0, 0)$. Its incoming ratio is $\phi$, with growth candidate $\sqrt{1+\phi} = \phi$, so its next step is $w_1 = 1$ by the departure calculation. No controller with a negative-index iterate is needed. The four controller verifications, together with the buffers and tail argument, prove inductively the entire actual orbit. It has first coordinate zero except for one point with first coordinate $c = 1/2$ in each transition, followed by an exact return to zero. All stepsizes are the finite positive values selected by (2). Every primal point is interior by (30). Its transformed normal coordinate increases strictly at every step; gradient injectivity excludes repeated primal points. Thus all Bregman denominators are positive and there is no stopping event. Every ambient probe is finite, since $h^*$ is finite $C^2$ on the entire plane.

Each update is also the genuine Bregman proximal minimizer. Its first-order condition is $\nabla h(X_{k+1}) = p_k - \gamma_{k+1} \nabla f(X_k)$. Fenchel duality and strict convexity of $h^*$ make this minimizer unique; there is no hidden projection or additional constraint in (1).

By (21), the spike and return times tend to $+\infty$. At both such endpoints $a' = \varepsilon' = 0$, so their $y$ coordinate is $-Q'(t) \to 0$. Consequently

$$X_{\text{spike},j} \longrightarrow (1/2, 0), \qquad X_{\text{return},j} \longrightarrow (0, 0).$$

The smooth ambient convex function $f(x, y) = x^2 + y$ is nonnegative on $K$ and vanishes there only at $(0, 0)$. These two subsequences prove all limiting assertions in Theorem 1. For every compact convex subset $J \subset C$, continuity and positive definiteness give $m_J = \min_J \lambda_{\min}(\nabla^2 h) > 0$. Hence any $L \geq 2/m_J$ makes $Lh - f$ convex on $J$. Along with $g = 0$ and the established kernel properties, this verifies the source Assumption 2.1. In particular the unique solution is allowed to be on the boundary of $C$, since the source problem minimizes on its closure.

## 6.1 Why stronger smoothness does not apply

At every centered transition start, $a' = \varepsilon' = 0$ and the second derivatives of the joins vanish. In transformed dual coordinates the Hessian is diagonal with entries $1/\varepsilon_j$ and $\lambda_j$. Thus at the corresponding primal point the $xx$ entry of $\nabla^2 h$ is $\varepsilon_j \to 0$, whereas $f_{xx} = 2$. A common relative-smoothness constant covering these bounded interior points would require

$$L\varepsilon_j \geq 2 \quad \text{for every } j,$$

which is impossible for finite $L$. The boundary-uniform smoothness hypothesis is therefore explicitly violated. Results that impose that stronger hypothesis are not contradicted.

For additional geometric perspective, $-\psi_t > 0$ and $\psi(s,t) \to |s|$ as $t \to +\infty$. Consequently $\psi(s,t) \geq |s|$ everywhere, and conjugating at $y = 0$ shows $h(x,0) = 0$ for every $|x| \leq 1$. The kernel has a flat bottom edge, while the objective restricted to that edge is strictly convex. This does not impair Legendre strict convexity, which is required on convex subsets of $\operatorname{dom} \partial h = C$.

## 6.2 The exact conclusion and its limits

This construction refutes full convergence of the undamped rule under the original weaker compact-interior local relative smoothness, even when the objective itself has a unique minimizer. It is a distinct assertion from an example with several objective minimizers. Theorem 2.7(ii) of the pinned source states boundedness and a unique optimal limit point. The present orbit has exactly that property: $(0,0)$ is its unique optimal cluster point, while $(1/2,0)$ is a nonoptimal cluster point. Thus the example does not contradict that theorem's wording. It addresses the stronger undamped full-convergence possibility discussed after Remark 2.8. No conclusion is asserted for the damped modification there, for interior-solution hypotheses, or for an uninspected journal version. The citation and controller throughout this paper refer only to the pinned arXiv version 2. No priority or novelty claim is intended.

# References

[1] H. Ou, P. Latafat, and A. Themelis. Linesearch-free adaptive Bregman proximal gradient for convex minimization under local relative smoothness. arXiv:2508.01353v2, 5 June 2026. `https://arxiv.org/abs/2508.01353v2`.